

Conjugate Directions and Conjugate Gradients

Learning objectives

By the end of this chapter, you should be able to:

- explain how conjugate-direction methods sit between steepest descent and Newton's method for a strongly convex quadratic;
- derive conjugacy from the requirement that each iterate minimize the objective over the span of all directions used so far;
- prove the expanding-subspace optimality and finite-termination properties of the conjugate-direction method;
- derive the conjugate-gradient recurrences and prove their residual, orthogonality, conjugacy, and Krylov-subspace properties;
- state and prove the condition-number convergence bound for conjugate gradients, and explain how PD preconditioning changes the relevant spectrum;
- explain why residual refresh and restart can recover accuracy lost to finite-precision arithmetic without improving exact-arithmetic linear CG; and
- distinguish linear conjugate gradients from nonlinear conjugate-gradient variants such as FR, PRP, HS, and DY.

The previous chapter derived steepest descent in two ways. Replacing the Hessian in a quadratic local model by the identity gives the direction $-\nabla f(\mathbf{x})$; the same direction, after normalization, gives the fastest instantaneous decrease measured in the Euclidean norm. Whenever the gradient is nonzero, this direction is descending. If f is L -smooth, then every step length $0 < \alpha \leq 1/L$ satisfies the Armijo inequality with parameter $\sigma = 1/2$. For a μ -strongly convex and L -smooth objective, the resulting linear rate is governed by the condition number L/μ , which measures how distorted the local landscape is. Without strong convexity, the rate may deteriorate to a sublinear one.

This chapter asks whether one can retain the inexpensive iterations of a first-order method while using curvature more effectively. Conjugate directions provide the geometric answer for strongly convex quadratics, and conjugate gradients (CG) provide a recursive way to generate those directions. The construction was developed for linear systems with symmetric PD coefficient matrices, is equivalent to minimizing a strongly convex quadratic, and later inspired nonlinear conjugate-gradient methods for general unconstrained optimization.

11.1 Between steepest descent and Newton's method

For the problem classes studied in Chapters 9 and 10, the main contrasts are as follows.

	Newton's method	Steepest descent
Information	Gradient and Hessian; a linear system is solved at every iteration.	Gradient only; one direction is obtained immediately.
Local speed	Quadratic under the usual local regularity assumptions; one full step for a strictly convex quadratic.	Generally linear on the strongly convex smooth class, and potentially sublinear without strong convexity.
Robustness	Pure Newton can be sensitive to the initial point, singular or indefinite Hessians, and full steps.	With an appropriate step rule, convergence theory applies from arbitrary initial points on broad smooth convex problem classes.
Geometry	Uses the complete local quadratic curvature.	Its rate can deteriorate severely with the condition number.

Thus neither method dominates the other. In particular, “global convergence” here means convergence from an arbitrary initial point under the stated assumptions; it does not mean that steepest descent finds a global minimizer of every nonconvex function. We seek a method whose iterations use only matrix–vector products and vector operations, but whose behavior on a quadratic is substantially better than steepest descent.

11.2 The quadratic problem and a two-direction question

Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be symmetric PD and let $\mathbf{c} \in \mathbb{R}^n$. Consider

$$\min_{\mathbf{x} \in \mathbb{R}^n} q(\mathbf{x}), \quad q(\mathbf{x}) := \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} - \mathbf{c}^\top \mathbf{x}. \quad (11.1)$$

Its unique minimizer $\mathbf{x}^*$ is characterized by $\mathbf{Q}\mathbf{x}^* = \mathbf{c}$. Throughout the linear-CG part of this chapter, write

$$\mathbf{g}^{(k)} := \nabla q(\mathbf{x}^{(k)}) = \mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}. \quad (11.2)$$

We call $\mathbf{g}^{(k)}$ the *gradient residual*. The conventional linear-system residual is $\mathbf{c} - \mathbf{Q}\mathbf{x}^{(k)} = -\mathbf{g}^{(k)}$.

Warning

Many numerical-linear-algebra references use $\mathbf{r}^{(k)} = \mathbf{c} - \mathbf{Q}\mathbf{x}^{(k)}$ and start with $\mathbf{d}^{(0)} = \mathbf{r}^{(0)}$. Here the residual is the gradient $\mathbf{g}^{(k)} = \mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}$, so the initial direction is $\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}$. Both conventions are valid, but their signs must not be mixed.

Lemma 11.1 (Exact line search on a strongly convex quadratic). *Let $\mathbf{x} \in \mathbb{R}^n$ and $\mathbf{d} \neq \mathbf{0}$. The function $\alpha \mapsto q(\mathbf{x} + \alpha \mathbf{d})$ has the unique minimizer*

$$\alpha_* = -\frac{\nabla q(\mathbf{x})^\top \mathbf{d}}{\mathbf{d}^\top \mathbf{Q} \mathbf{d}}. \quad (11.3)$$

At the new point $\mathbf{x}_+ = \mathbf{x} + \alpha_ \mathbf{d}$, $\nabla q(\mathbf{x}_+)^\top \mathbf{d} = 0$.*

Proof. Expansion along the line gives

$$q(\mathbf{x} + \alpha \mathbf{d}) = q(\mathbf{x}) + \alpha \nabla q(\mathbf{x})^\top \mathbf{d} + \frac{\alpha^2}{2} \mathbf{d}^\top \mathbf{Q} \mathbf{d}.$$

Since $\mathbf{Q} \succ 0$ and $\mathbf{d} \neq \mathbf{0}$, the coefficient $\mathbf{d}^\top \mathbf{Q} \mathbf{d}$ is positive. The line objective is therefore strictly convex, and its derivative vanishes uniquely at Equation (11.3). Moreover,

$$\nabla q(\mathbf{x}_+) = \nabla q(\mathbf{x}) + \alpha_* \mathbf{Q} \mathbf{d},$$

so multiplication by $\mathbf{d}^\top$ gives the stated orthogonality. $\square$

The desired “memory” property is best seen from the first two directions. Start from $\mathbf{x}^{(0)}$ and perform an exact line search along a nonzero direction $\mathbf{d}^{(0)}$:

$$\mathbf{x}^{(1)} = \mathbf{x}^{(0)} + \alpha^{(0)} \mathbf{d}^{(0)}, \quad (\mathbf{g}^{(1)})^\top \mathbf{d}^{(0)} = 0.$$

Choose a second direction $\mathbf{d}^{(1)}$ that is linearly independent of $\mathbf{d}^{(0)}$, perform another exact line search, and set

$$\mathbf{x}^{(2)} = \mathbf{x}^{(1)} + \alpha^{(1)} \mathbf{d}^{(1)}, \quad (\mathbf{g}^{(2)})^\top \mathbf{d}^{(1)} = 0.$$

We want more than line optimality: $\mathbf{x}^{(2)}$ should minimize q over the entire affine plane

$$\mathcal{S}_2 = \mathbf{x}^{(0)} + \text{span}\{\mathbf{d}^{(0)}, \mathbf{d}^{(1)}\}.$$

Lemma 11.2 (Optimality over an affine subspace). *Let $\mathcal{V}$ be a linear subspace of $\mathbb{R}^n$ and let $\bar{\mathbf{x}} \in \mathbf{x}^{(0)} + \mathcal{V}$. Then $\bar{\mathbf{x}}$ is the unique minimizer of q over $\mathbf{x}^{(0)} + \mathcal{V}$ if and only if*

$$\nabla q(\bar{\mathbf{x}})^\top \mathbf{v} = 0 \quad \text{for every } \mathbf{v} \in \mathcal{V}. \quad (11.4)$$

Proof. If $\bar{\mathbf{x}}$ is a minimizer, then for every $\mathbf{v} \in \mathcal{V}$ the differentiable one-dimensional function $t \mapsto q(\bar{\mathbf{x}} + t\mathbf{v})$ is minimized at $t = 0$. Its derivative there is $\nabla q(\bar{\mathbf{x}})^\top \mathbf{v}$, proving necessity.

Conversely, suppose Equation (11.4) holds. For every $\mathbf{v} \in \mathcal{V}$,

$$q(\bar{\mathbf{x}} + \mathbf{v}) - q(\bar{\mathbf{x}}) = \nabla q(\bar{\mathbf{x}})^\top \mathbf{v} + \frac{1}{2} \mathbf{v}^\top \mathbf{Q} \mathbf{v} = \frac{1}{2} \mathbf{v}^\top \mathbf{Q} \mathbf{v}.$$

This quantity is positive whenever $\mathbf{v} \neq \mathbf{0}$, because $\mathbf{Q} \succ 0$. Hence $\bar{\mathbf{x}}$ is the unique minimizer over the affine subspace. $\square$

By Lemma 11.2, the two required equations are $(\mathbf{g}^{(2)})^\top \mathbf{d}^{(0)} = 0$ and $(\mathbf{g}^{(2)})^\top \mathbf{d}^{(1)} = 0$. The second is supplied automatically by the second exact line search. For the first, use

$$\mathbf{g}^{(2)} = \mathbf{g}^{(1)} + \alpha^{(1)} \mathbf{Q} \mathbf{d}^{(1)}$$

and $(\mathbf{g}^{(1)})^\top \mathbf{d}^{(0)} = 0$ to obtain

$$(\mathbf{g}^{(2)})^\top \mathbf{d}^{(0)} = \alpha^{(1)} (\mathbf{d}^{(1)})^\top \mathbf{Q} \mathbf{d}^{(0)}. \quad (11.5)$$

For a nontrivial second step, $\alpha^{(1)} \neq 0$, preserving the first line optimality condition is therefore equivalent to

$$(\mathbf{d}^{(1)})^\top \mathbf{Q} \mathbf{d}^{(0)} = 0.$$

This is the conjugacy condition.

Warning

The assumption $\alpha^{(1)} \neq 0$ matters in the preceding implication. If $\alpha^{(1)} = 0$, then $\mathbf{x}^{(2)} = \mathbf{x}^{(1)}$ and Equation (11.5) gives no information about $(\mathbf{d}^{(1)})^\top \mathbf{Q} \mathbf{d}^{(0)}$. This harmless degenerate case occurs when the enlarged subspace is already minimized at $\mathbf{x}^{(1)}$.

11.3 Conjugate directions: retaining all previous progress

Definition 11.1 ($\mathbf{Q}$ -conjugate directions). Nonzero vectors $\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(m-1)}$ are *conjugate with respect to $\mathbf{Q}$* if

$$(\mathbf{d}^{(i)})^\top \mathbf{Q} \mathbf{d}^{(j)} = 0 \quad \text{whenever } i \neq j. \quad (11.6)$$

Since $\mathbf{Q} \succ 0$, the formula $\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{Q}} := \mathbf{u}^\top \mathbf{Q} \mathbf{v}$ defines an inner product. Thus conjugacy is orthogonality in the $\mathbf{Q}$ -geometry.

Warning

$\mathbf{Q}$ -conjugacy is not generally Euclidean orthogonality. The condition $\mathbf{u}^\top \mathbf{Q} \mathbf{v} = 0$ need not imply $\mathbf{u}^\top \mathbf{v} = 0$. Only when $\mathbf{Q}$ is a positive multiple of the identity do the two notions coincide.

Lemma 11.3 (Independence of conjugate directions). *Every collection of nonzero $\mathbf{Q}$ -conjugate vectors is linearly independent. Consequently, there are at most n such vectors in $\mathbb{R}^n$.*

Proof. Suppose $\sum_{i=0}^{m-1} \gamma_i \mathbf{d}^{(i)} = \mathbf{0}$. Multiplying by $(\mathbf{d}^{(j)})^\top \mathbf{Q}$ and using conjugacy yields

$$\gamma_j (\mathbf{d}^{(j)})^\top \mathbf{Q} \mathbf{d}^{(j)} = 0.$$

The second factor is positive, so $\gamma_j = 0$ for every j . □

Suppose m nonzero $\mathbf{Q}$ -conjugate directions are available. The conjugate-direction method performs one exact line search along each:

$$\alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^\top \mathbf{Q} \mathbf{d}^{(k)}}, \quad \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}. \quad (11.7)$$

The gradient residual obeys the inexpensive recursion

$$\mathbf{g}^{(k+1)} = \mathbf{g}^{(k)} + \alpha^{(k)} \mathbf{Q} \mathbf{d}^{(k)}. \quad (11.8)$$

Algorithm: Conjugate-direction method

Given $\mathbf{x}^{(0)}$ and nonzero $\mathbf{Q}$ -conjugate directions $\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(m-1)}$, set $\mathbf{g}^{(0)} = \mathbf{Q} \mathbf{x}^{(0)} - \mathbf{c}$. For $k = 0, \dots, m-1$:

1. Compute $\alpha^{(k)}$ from Equation (11.7).
2. Set $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}$.

3. Set $\mathbf{g}^{(k+1)} = \mathbf{g}^{(k)} + \alpha^{(k)} \mathbf{Q} \mathbf{d}^{(k)}$.

Theorem 11.4 (Memory and expanding-subspace optimality). *For every $k = 0, 1, \dots, m$, the conjugate-direction iterate satisfies*

$$(\mathbf{g}^{(k)})^\top \mathbf{d}^{(i)} = 0, \quad i = 0, \dots, k-1, \quad (11.9)$$

and is the unique minimizer of q over

$$\mathbf{x}^{(0)} + \text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(k-1)}\}. \quad (11.10)$$

Proof. The claim for $k = 0$ is vacuous, and $\mathbf{x}^{(0)}$ is the unique point of the zero-dimensional affine space. Suppose the orthogonality statement holds at iteration k . Exact line search gives $(\mathbf{g}^{(k+1)})^\top \mathbf{d}^{(k)} = 0$. For every $i < k$,

$$\begin{aligned} (\mathbf{g}^{(k+1)})^\top \mathbf{d}^{(i)} &= (\mathbf{g}^{(k)})^\top \mathbf{d}^{(i)} + \alpha^{(k)} (\mathbf{d}^{(k)})^\top \mathbf{Q} \mathbf{d}^{(i)} \\ &= 0, \end{aligned}$$

where the two terms vanish by the induction hypothesis and conjugacy, respectively. This proves Equation (11.9) by induction.

The update formula places $\mathbf{x}^{(k)}$ in the affine space Equation (11.10). Its gradient is orthogonal to every generator of that space by Equation (11.9). The conclusion now follows from Lemma 11.2. $\square$

Corollary 11.5 (Finite termination). *If $m = n$, the conjugate-direction method reaches $\mathbf{x}^*$ in at most n iterations in exact arithmetic.*

Proof. By Lemma 11.3, the n directions form a basis of $\mathbb{R}^n$. Hence the affine space in Equation (11.10) with $k = n$ is all of $\mathbb{R}^n$. The unique minimizer identified by Theorem 11.4 is therefore $\mathbf{x}^*$. $\square$

The precise intermediate iterates depend on the chosen conjugate basis and on its order, even though every complete conjugate basis reaches the same answer in at most n steps. A basis whose early spans contain a useful error component can perform much better before termination than another basis. Scaling a direction by a nonzero constant, however, does not change its search line and therefore does not change the next exact-line-search iterate.

For a concrete example used throughout this chapter, take

$$q(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x}, \quad \mathbf{Q} = \begin{pmatrix} 5 & -4 \\ -4 & 5 \end{pmatrix}, \quad \mathbf{c} = \mathbf{0}, \quad \mathbf{x}^{(0)} = \begin{pmatrix} 1 \\ 0.8 \end{pmatrix}. \quad (11.11)$$

The eigenvalues of $\mathbf{Q}$ are 1 and 9, so $\kappa(\mathbf{Q}) = 9$, $\mathbf{x}^* = \mathbf{0}$, and $q(\mathbf{x}^{(0)}) = 0.9$. Two ordered conjugate bases are

$$\text{CD-A: } \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix}, \quad \text{CD-B: } \begin{pmatrix} -1 \\ -2 \end{pmatrix}, \begin{pmatrix} -2 \\ -1 \end{pmatrix}.$$

Exact line searches give the objective sequences 0.9, 0.09, 0 for CD-A and 0.9, 0.72, 0 for CD-B. See the trajectories in Figure 11.1. By comparison, exact-search steepest descent begins with 0.9, 0.576, 0.36864 and continues beyond the second update.

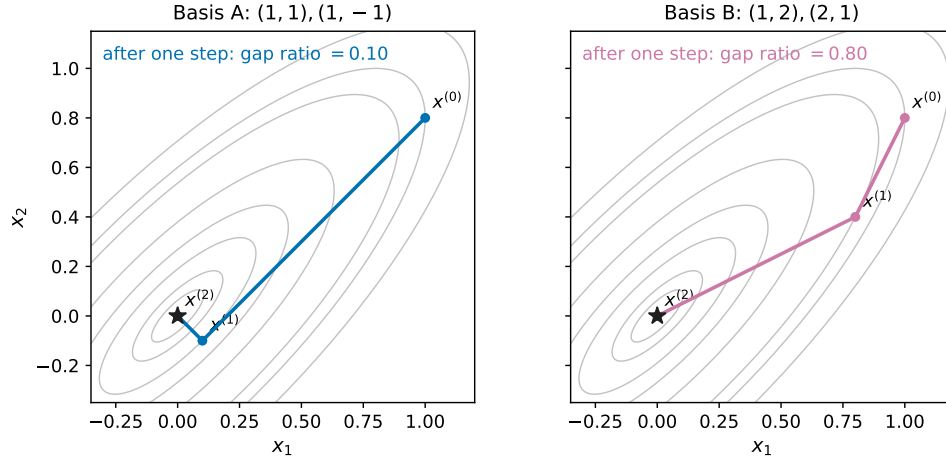


Figure 11.1: Exact line searches along two different $\mathbf{Q}$ -conjugate bases, compared in the same contour plot with exact-search steepest descent. Both conjugate bases terminate in two exact-arithmetic updates, whereas steepest descent continues to cross the narrow valley.

11.4 Conjugate gradients: generating the directions recursively

The conjugate-direction method has a serious practical weakness: it assumes that an entire conjugate basis is already known. An attractive construction should instead be

- recursive, using only information from the current and preceding iterations;
- inexpensive, requiring no matrix factorization and no storage of all earlier directions;
- simple, with one multiplication by $\mathbf{Q}$ per iteration; and
- exact in the idealized sense of retaining mutual conjugacy and finite termination.

CG satisfies all four requirements in exact arithmetic.

Set

$$\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}.$$

For $k \geq 1$, combine the new negative gradient with only the previous direction:

$$\mathbf{d}^{(k)} = -\mathbf{g}^{(k)} + \beta^{(k)} \mathbf{d}^{(k-1)}. \quad (11.12)$$

Requiring $\mathbf{d}^{(k)}$ to be conjugate to $\mathbf{d}^{(k-1)}$ gives

$$\beta^{(k)} = \frac{(\mathbf{g}^{(k)})^\top \mathbf{Q} \mathbf{d}^{(k-1)}}{(\mathbf{d}^{(k-1)})^\top \mathbf{Q} \mathbf{d}^{(k-1)}}. \quad (11.13)$$

It is not yet evident that conjugacy with the immediately preceding direction implies conjugacy with every earlier direction. The Krylov structure supplies the missing argument.

Definition 11.2 (Krylov subspace). For $j \geq 1$, define

$$\mathcal{K}_j(\mathbf{Q}, \mathbf{g}^{(0)}) := \text{span}\{\mathbf{g}^{(0)}, \mathbf{Q}\mathbf{g}^{(0)}, \dots, \mathbf{Q}^{j-1}\mathbf{g}^{(0)}\}, \quad (11.14)$$

and set $\mathcal{K}_0(\mathbf{Q}, \mathbf{g}^{(0)}) := \{\mathbf{0}\}$.

For the moment, define each step length by exact line search,

$$\alpha^{(k)} = -\frac{(\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)}}{(\mathbf{d}^{(k)})^\top \mathbf{Q} \mathbf{d}^{(k)}}, \quad (11.15)$$

and update $\mathbf{x}^{(k+1)}$ and $\mathbf{g}^{(k+1)}$ by Equations (11.7) and (11.8). If the new gradient is nonzero, define $\beta^{(k+1)}$ by Equation (11.13) and then use Equation (11.12).

Theorem 11.6 (The CG recursion package). *In exact arithmetic, the CG iterates and all directions generated before termination have the following properties.*

1. All generated directions are mutually $\mathbf{Q}$ -conjugate.
2. Whenever $\mathbf{d}^{(k)}$ is generated,

$$\text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(k)}\} = \text{span}\{\mathbf{g}^{(0)}, \dots, \mathbf{g}^{(k)}\} = \mathcal{K}_{k+1}(\mathbf{Q}, \mathbf{g}^{(0)}). \quad (11.16)$$

3. Every generated iterate $\mathbf{x}^{(k)}$, including a terminal iterate, uniquely minimizes q over $\mathbf{x}^{(0)} + \mathcal{K}_k(\mathbf{Q}, \mathbf{g}^{(0)})$, and

$$\mathbf{g}^{(k)} \perp \mathcal{K}_k(\mathbf{Q}, \mathbf{g}^{(0)}). \quad (11.17)$$

4. The generated gradient residuals are mutually Euclidean-orthogonal: $(\mathbf{g}^{(i)})^\top \mathbf{g}^{(j)} = 0$ whenever $i \neq j$.

5. At every nonterminal iteration the line-search coefficient simplifies as below; whenever the next direction is generated, so does the conjugacy coefficient:

$$\alpha^{(k)} = \frac{(\mathbf{g}^{(k)})^\top \mathbf{g}^{(k)}}{(\mathbf{d}^{(k)})^\top \mathbf{Q} \mathbf{d}^{(k)}}, \quad (11.18)$$

$$\beta^{(k+1)} = \frac{(\mathbf{g}^{(k+1)})^\top \mathbf{g}^{(k+1)}}{(\mathbf{g}^{(k)})^\top \mathbf{g}^{(k)}}. \quad (11.19)$$

Proof. We prove the statements together by induction. Let

$$\mathcal{V}_j := \text{span}\{\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(j-1)}\}, \quad \mathcal{V}_0 := \{\mathbf{0}\}.$$

Initially, $\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}$, so $\mathcal{V}_1 = \mathcal{K}_1(\mathbf{Q}, \mathbf{g}^{(0)})$. Also $\mathbf{x}^{(0)}$ is the unique point of $\mathbf{x}^{(0)} + \mathcal{K}_0$, and $\mathbf{g}^{(0)} \perp \mathcal{K}_0$.

At the beginning of a nonterminal iteration k , use the following induction invariants: the directions through $\mathbf{d}^{(k)}$ are mutually conjugate, $\mathcal{V}_{k+1} = \mathcal{K}_{k+1}$, the iterate $\mathbf{x}^{(k)}$ satisfies the Galerkin condition over $\mathcal{K}_k$, and the residuals through $\mathbf{g}^{(k)}$ are mutually orthogonal. The preceding base observations verify these invariants for $k = 0$. For $k = 0$, $(\mathbf{g}^{(0)})^\top \mathbf{d}^{(0)} = -\|\mathbf{g}^{(0)}\|^2$. For $k \geq 1$, Equation (11.12) and $\mathbf{d}^{(k-1)} \in \mathcal{K}_k$ together with Equation (11.17) give

$$(\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)} = -\|\mathbf{g}^{(k)}\|^2. \quad (11.20)$$

Thus $\mathbf{d}^{(k)}$ is descending, $\alpha^{(k)} > 0$, and substitution into Equation (11.15) proves Equation (11.18).

The directions available through $\mathbf{d}^{(k)}$ are mutually conjugate by the induction hypothesis. Applying Theorem 11.4 to these first $k+1$ directions shows that $\mathbf{x}^{(k+1)}$ uniquely minimizes q over $\mathbf{x}^{(0)} + \mathcal{V}_{k+1}$ and that

$$\mathbf{g}^{(k+1)} \perp \mathcal{V}_{k+1}. \quad (11.21)$$

Because $\mathcal{V}_{k+1} = \mathcal{K}_{k+1}$ by induction, this proves the next Galerkin condition. Every earlier gradient belongs to $\mathcal{K}_{k+1}$, so Equation (11.21) also proves $(\mathbf{g}^{(k+1)})^\top \mathbf{g}^{(i)} = 0$ for $i = 0, \dots, k$.

The gradient recursion and the inclusion $\mathbf{d}^{(k)} \in \mathcal{K}_{k+1}$ imply

$$\mathbf{g}^{(k+1)} = \mathbf{g}^{(k)} + \alpha^{(k)} \mathbf{Q} \mathbf{d}^{(k)} \in \mathcal{K}_{k+2}.$$

If $\mathbf{g}^{(k+1)} \neq \mathbf{0}$, its orthogonality to $\mathcal{K}_{k+1}$ shows that it is not in that subspace. By the induction hypothesis and Lemma 11.3, $\dim \mathcal{K}_{k+1} = k + 1$. Hence the span of $\mathcal{K}_{k+1}$ and $\mathbf{g}^{(k+1)}$ has dimension $k + 2$. It is contained in $\mathcal{K}_{k+2}$, which has at most $k + 2$ generators; therefore

$$\mathcal{K}_{k+2} = \text{span}(\mathcal{K}_{k+1}, \mathbf{g}^{(k+1)}).$$

Since $\mathbf{d}^{(k+1)} = -\mathbf{g}^{(k+1)} + \beta^{(k+1)} \mathbf{d}^{(k)}$, adding $\mathbf{d}^{(k+1)}$ to $\mathcal{V}_{k+1}$ produces the same enlarged span. This proves both span equalities at index $k + 1$.

It remains to prove that $\mathbf{d}^{(k+1)}$ is conjugate to all previous directions. Its conjugacy with $\mathbf{d}^{(k)}$ follows directly from Equation (11.13). If $i < k$, the gradient recursion gives

$$\mathbf{Q} \mathbf{d}^{(i)} = \frac{\mathbf{g}^{(i+1)} - \mathbf{g}^{(i)}}{\alpha^{(i)}} \in \mathcal{K}_{k+1}. \quad (11.22)$$

Therefore $(\mathbf{g}^{(k+1)})^\top \mathbf{Q} \mathbf{d}^{(i)} = 0$ by Equation (11.21). Using the direction recurrence and the induction hypothesis,

$$\begin{aligned} (\mathbf{d}^{(k+1)})^\top \mathbf{Q} \mathbf{d}^{(i)} &= -(\mathbf{g}^{(k+1)})^\top \mathbf{Q} \mathbf{d}^{(i)} + \beta^{(k+1)} (\mathbf{d}^{(k)})^\top \mathbf{Q} \mathbf{d}^{(i)} \\ &= 0. \end{aligned}$$

Thus mutual conjugacy is preserved.

Finally, use $\mathbf{Q} \mathbf{d}^{(k)} = (\mathbf{g}^{(k+1)} - \mathbf{g}^{(k)})/\alpha^{(k)}$ and the newly proved gradient orthogonality to obtain

$$(\mathbf{g}^{(k+1)})^\top \mathbf{Q} \mathbf{d}^{(k)} = \frac{\|\mathbf{g}^{(k+1)}\|^2}{\alpha^{(k)}}.$$

Equation (11.18) gives $(\mathbf{d}^{(k)})^\top \mathbf{Q} \mathbf{d}^{(k)} = \|\mathbf{g}^{(k)}\|^2 / \alpha^{(k)}$. Their quotient is precisely Equation (11.19). This completes the induction. $\square$

The rightmost space in Equation (11.16) is the order- $(k+1)$ Krylov space generated by $\mathbf{Q}$ and $\mathbf{g}^{(0)}$. Thus CG does not search an arbitrary expanding subspace: at iteration k , it chooses the best point in the space generated by repeatedly applying $\mathbf{Q}$ to the initial gradient.

Corollary 11.7 (Finite termination of CG). *CG reaches $\mathbf{x}^*$ in at most n iterations in exact arithmetic.*

Proof. If CG has not yet terminated, its generated directions are nonzero and mutually $\mathbf{Q}$ -conjugate by Theorem 11.6; hence they are linearly independent by Lemma 11.3. At most n such directions can exist in $\mathbb{R}^n$. If n directions are generated, Corollary 11.5 gives $\mathbf{x}^{(n)} = \mathbf{x}^*$. $\square$

11.4.1 The implementable CG algorithm

Both the unsimplified and simplified coefficients require only one multiplication by $\mathbf{Q}$ per iteration, provided that $\mathbf{w}^{(k)} = \mathbf{Q}\mathbf{d}^{(k)}$ is stored and reused in the denominator and the gradient update. The simplification instead reduces the number of inner products. To make this precise, cache

$$\gamma^{(k)} = (\mathbf{g}^{(k)})^\top \mathbf{g}^{(k)}, \quad \delta^{(k)} = (\mathbf{d}^{(k)})^\top \mathbf{w}^{(k)}.$$

Count one completed iteration and include the computation of $\gamma^{(k+1)}$, which also supplies the next gradient-norm stopping test. The unsimplified formulas require the four inner products

$$\delta^{(k)}, \quad (\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)}, \quad (\mathbf{g}^{(k+1)})^\top \mathbf{w}^{(k)}, \quad \gamma^{(k+1)},$$

whereas the simplified formulas require only $\delta^{(k)}$ and $\gamma^{(k+1)}$. Thus both forms use one $\mathbf{Q}$ -vector product and the same three vector updates, while simplification reduces four length- n inner products to two. The initial $\gamma^{(0)}$ is a one-time computation outside this count. In a straightforward distributed implementation this also reduces four global reductions to two; if independent products are batched, both forms can use two synchronization stages, but the simplified form still performs fewer local inner products and reduces fewer scalars.

Algorithm: Conjugate gradient for a PD quadratic

Given $\mathbf{x}^{(0)}$, set $\mathbf{g}^{(0)} = \mathbf{Q}\mathbf{x}^{(0)} - \mathbf{c}$ and $\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}$. For $k = 0, 1, 2, \dots$ while $\mathbf{g}^{(k)} \neq \mathbf{0}$:

1. Set $\mathbf{w}^{(k)} = \mathbf{Q}\mathbf{d}^{(k)}$ and

$$\alpha^{(k)} = \frac{(\mathbf{g}^{(k)})^\top \mathbf{g}^{(k)}}{(\mathbf{d}^{(k)})^\top \mathbf{w}^{(k)}}.$$

2. Set $\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)}\mathbf{d}^{(k)}$ and $\mathbf{g}^{(k+1)} = \mathbf{g}^{(k)} + \alpha^{(k)}\mathbf{w}^{(k)}$.

3. If $\mathbf{g}^{(k+1)} = \mathbf{0}$, stop. Otherwise set

$$\beta^{(k+1)} = \frac{(\mathbf{g}^{(k+1)})^\top \mathbf{g}^{(k+1)}}{(\mathbf{g}^{(k)})^\top \mathbf{g}^{(k)}}, \quad \mathbf{d}^{(k+1)} = -\mathbf{g}^{(k+1)} + \beta^{(k+1)}\mathbf{d}^{(k)}.$$

Apart from the multiplication $\mathbf{w}^{(k)} = \mathbf{Q}\mathbf{d}^{(k)}$, each iteration uses only inner products and vector updates. For a sparse matrix this product costs $O(\text{nnz}(\mathbf{Q}))$, and the method stores only a fixed number of length- n vectors in addition to the representation of $\mathbf{Q}$. No matrix factorization or growing list of past directions is required.

The exact-arithmetic identities explain the design, but floating-point roundoff gradually destroys perfect orthogonality and conjugacy. In an implementation, the recursively updated gradient residual should occasionally be checked against the freshly evaluated quantity $\mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}$. [Section 11.8](#) gives a concrete residual-refresh experiment. A typical stopping condition is

$$\|\mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}\| \leq \varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \|\mathbf{c}\|. \quad (11.23)$$

Finite termination in at most n iterations is not guaranteed in floating point arithmetic.

11.5 A common quadratic comparison

For the common quadratic [Equation \(11.11\)](#), the objective values are

method	$q(\mathbf{x}^{(0)})$	$q(\mathbf{x}^{(1)})$	$q(\mathbf{x}^{(2)})$
exact-search steepest descent	0.9	0.576	0.36864
CD-A	0.9	0.09	0
CD-B	0.9	0.72	0
CG	0.9	0.576	0

In particular, CG and steepest descent share their first step, while CG reaches the solution on its second step. The full convergence trajectories and convergence behavior are documented in Figure 11.2.

The expanding-subspace viewpoint explains the contrast among steepest descent, arbitrary conjugate directions, and CG. Exact-search steepest descent minimizes over only the current line and may undo the geometry of earlier steps. A preselected conjugate basis retains every earlier line optimality condition, but its early subspaces depend on the basis. CG chooses the specific Krylov spaces and recursively constructs a conjugate basis for them.

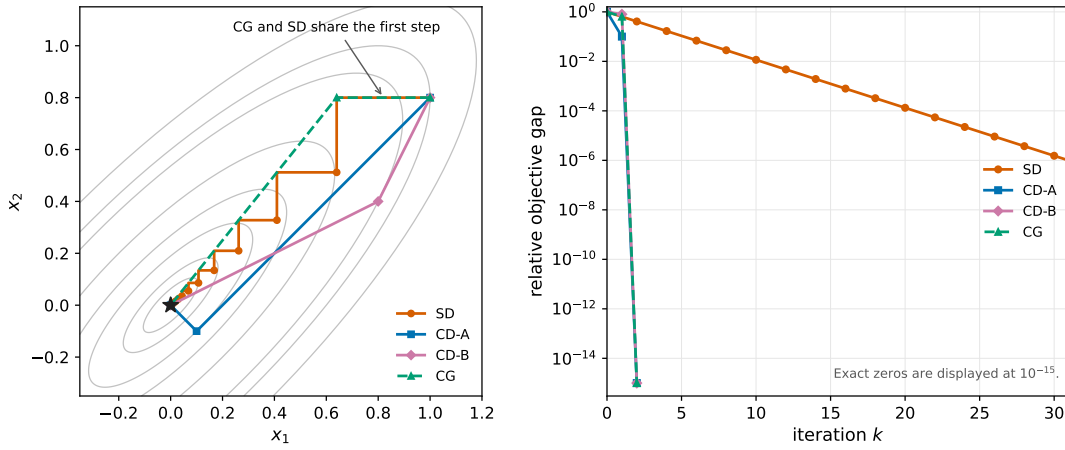


Figure 11.2: Trajectories and convergence histories of steepest descent, two conjugate-direction choices, and CG on the same strongly convex quadratic. All conjugate-direction choices terminate finitely in exact arithmetic, but their intermediate behavior can differ.

The figure illustrates a point that the finite-termination theorem alone does not capture: when n is large and only a small number of iterations is affordable, the quality of the successive subspaces matters. CG's Krylov choice leads to a useful quantitative bound before finite termination.

11.6 Convergence before finite termination

Define the error and the Q -norm by

$$\mathbf{e}^{(k)} := \mathbf{x}^{(k)} - \mathbf{x}^*, \quad \|\mathbf{v}\|_Q := \sqrt{\mathbf{v}^\top \mathbf{Q} \mathbf{v}}.$$

Completing the square gives

$$q(\mathbf{x}) - q(\mathbf{x}^*) = \frac{1}{2} \|\mathbf{x} - \mathbf{x}^*\|_Q^2. \quad (11.24)$$

Lemma 11.8 (Polynomial optimality of CG). *Let $\mathcal{P}_k$ be the polynomials of degree at most k . Before or at termination,*

$$\|e^{(k)}\|_Q = \min_{\substack{p \in \mathcal{P}_k \\ p(0)=1}} \|p(Q)e^{(0)}\|_Q. \quad (11.25)$$

Proof. Because $g^{(0)} = Qe^{(0)}$,

$$\mathcal{K}_k(Q, g^{(0)}) = \text{span}\{Qe^{(0)}, \dots, Q^k e^{(0)}\}.$$

Every vector of the form $e^{(0)} + z$ with z in this Krylov space can therefore be written as $p(Q)e^{(0)}$ for a polynomial $p \in \mathcal{P}_k$ satisfying $p(0) = 1$, and conversely. By [Theorem 11.6](#), $x^{(k)}$ minimizes q over $x^{(0)} + \mathcal{K}_k$. Equation [Equation \(11.24\)](#) converts this objective minimization into the Q -norm minimization in [Equation \(11.25\)](#). $\square$

Theorem 11.9 (CG condition-number bound). *Let*

$$\kappa := \kappa(Q) = \frac{\lambda_{\max}(Q)}{\lambda_{\min}(Q)}.$$

If $\kappa = 1$, CG reaches x^ in at most one iteration. If $\kappa > 1$, then every generated iterate, including the terminal one, satisfies*

$$\|e^{(k)}\|_Q \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k \|e^{(0)}\|_Q. \quad (11.26)$$

Consequently,

$$q(x^{(k)}) - q(x^*) \leq 4 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^{2k} (q(x^{(0)}) - q(x^*)). \quad (11.27)$$

Proof. Write $\mu = \lambda_{\min}(Q)$ and $L = \lambda_{\max}(Q)$. The spectral theorem and [Lemma 11.8](#) imply that, for every $p \in \mathcal{P}_k$ with $p(0) = 1$,

$$\|e^{(k)}\|_Q \leq \max_{\lambda \in [\mu, L]} |p(\lambda)| \|e^{(0)}\|_Q. \quad (11.28)$$

Indeed, if $Q = U \text{diag}(\lambda_1, \dots, \lambda_n) U^T$ and $U^T e^{(0)} = (a_1, \dots, a_n)^T$, then

$$\|p(Q)e^{(0)}\|_Q^2 = \sum_{i=1}^n \lambda_i p(\lambda_i)^2 a_i^2 \leq \left(\max_{\lambda \in [\mu, L]} |p(\lambda)|^2 \right) \|e^{(0)}\|_Q^2.$$

If $L = \mu$, then $Q = \mu I$. Formula [Equation \(11.18\)](#) gives $\alpha^{(0)} = 1/\mu$ whenever the initial gradient is nonzero, and the first update yields $x^{(1)} = x^*$.

Suppose $L > \mu$. Let T_k be the degree- k Chebyshev polynomial of the first kind, characterized by $T_k(\cos \theta) = \cos(k\theta)$. Define

$$p_k(\lambda) = \frac{T_k((L + \mu - 2\lambda)/(L - \mu))}{T_k((L + \mu)/(L - \mu))}. \quad (11.29)$$

This polynomial has degree at most k and satisfies $p_k(0) = 1$. For $\lambda \in [\mu, L]$, the numerator argument lies in $[-1, 1]$, where $|T_k(t)| \leq 1$. Therefore

$$\max_{\lambda \in [\mu, L]} |p_k(\lambda)| \leq \frac{1}{|T_k((L + \mu)/(L - \mu))|}. \quad (11.30)$$

Set

$$\tau := \frac{\sqrt{L} + \sqrt{\mu}}{\sqrt{L} - \sqrt{\mu}} > 1, \quad \rho := \tau^{-1} = \frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1}.$$

A direct calculation gives

$$\frac{L + \mu}{L - \mu} = \frac{\tau + \tau^{-1}}{2}.$$

The Chebyshev recurrence $T_{j+1}(t) = 2tT_j(t) - T_{j-1}(t)$, with $T_0(t) = 1$ and $T_1(t) = t$, proves by induction that

$$T_k\left(\frac{\tau + \tau^{-1}}{2}\right) = \frac{\tau^k + \tau^{-k}}{2} \geq \frac{\tau^k}{2}.$$

Combining this estimate with Equations (11.28) and (11.30) proves Equation (11.26). Squaring that inequality and using Equation (11.24) proves Equation (11.27). $\square$

For exact-search steepest descent on the same quadratic, the corresponding worst-case $\mathbf{Q}$ -norm factor is $(\kappa - 1)/(\kappa + 1)$. CG replaces the dependence on κ by one on $\sqrt{\kappa}$, up to the leading constant 2. Thus the bounds suggest $O(\kappa \log(1/\varepsilon))$ iterations for steepest descent and $O(\sqrt{\kappa} \log(1/\varepsilon))$ for CG. These are worst-case estimates, not exact iteration-by-iteration ratios. In particular, clustered eigenvalues can make CG converge much faster than the condition number alone predicts.

11.7 Preconditioned conjugate gradients

As in preconditioned steepest descent, one can change the geometry so that the effective quadratic is better conditioned. Let $\mathbf{B}$ be a fixed symmetric PD matrix that approximates $\mathbf{Q}$ but is substantially cheaper to solve with.

Proposition 11.10 (Symmetric preconditioning and PCG). *Define*

$$\tilde{\mathbf{Q}} := \mathbf{B}^{-1/2} \mathbf{Q} \mathbf{B}^{-1/2}, \quad \mathbf{u} := \mathbf{B}^{1/2} \mathbf{x}, \quad \tilde{\mathbf{c}} := \mathbf{B}^{-1/2} \mathbf{c}. \quad (11.31)$$

Then $\mathbf{Q}\mathbf{x} = \mathbf{c}$ is equivalent to $\tilde{\mathbf{Q}}\mathbf{u} = \tilde{\mathbf{c}}$, and $\tilde{\mathbf{Q}}$ is symmetric PD. Ordinary CG on the transformed system is equivalent, in the original coordinates, to the iteration

$$\mathbf{z}^{(k)} = \mathbf{B}^{-1} \mathbf{g}^{(k)}, \quad (11.32)$$

$$\alpha^{(k)} = \frac{(\mathbf{g}^{(k)})^\top \mathbf{z}^{(k)}}{(\mathbf{d}^{(k)})^\top \mathbf{Q} \mathbf{d}^{(k)}}, \quad (11.33)$$

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}, \quad \mathbf{g}^{(k+1)} = \mathbf{g}^{(k)} + \alpha^{(k)} \mathbf{Q} \mathbf{d}^{(k)}, \quad (11.34)$$

$$\beta^{(k+1)} = \frac{(\mathbf{g}^{(k+1)})^\top \mathbf{z}^{(k+1)}}{(\mathbf{g}^{(k)})^\top \mathbf{z}^{(k)}}, \quad (11.35)$$

$$\mathbf{d}^{(0)} = -\mathbf{z}^{(0)}, \quad \mathbf{d}^{(k+1)} = -\mathbf{z}^{(k+1)} + \beta^{(k+1)} \mathbf{d}^{(k)}. \quad (11.36)$$

Let

$$\kappa_{\text{eff}}(\mathbf{B}) := \kappa(\tilde{\mathbf{Q}}) = \kappa(\mathbf{B}^{-1/2} \mathbf{Q} \mathbf{B}^{-1/2}).$$

If $\kappa_{\text{eff}}(\mathbf{B}) = 1$, PCG reaches $\mathbf{x}^*$ in at most one iteration. If $\kappa_{\text{eff}}(\mathbf{B}) > 1$, every generated iterate, including the terminal one, satisfies

$$\|\mathbf{e}^{(k)}\|_{\mathbf{Q}} \leq 2 \left(\frac{\sqrt{\kappa_{\text{eff}}(\mathbf{B})} - 1}{\sqrt{\kappa_{\text{eff}}(\mathbf{B})} + 1} \right)^k \|\mathbf{e}^{(0)}\|_{\mathbf{Q}}. \quad (11.37)$$

Proof. The PD square root $\mathbf{B}^{1/2}$ is invertible and $\mathbf{x} = \mathbf{B}^{-1/2}\mathbf{u}$. Substitution into $\mathbf{Q}\mathbf{x} = \mathbf{c}$ and multiplication by $\mathbf{B}^{-1/2}$ give $\tilde{\mathbf{Q}}\mathbf{u} = \tilde{\mathbf{c}}$. The matrix $\tilde{\mathbf{Q}}$ is symmetric, and for every nonzero $\mathbf{v}$,

$$\mathbf{v}^\top \tilde{\mathbf{Q}} \mathbf{v} = (\mathbf{B}^{-1/2}\mathbf{v})^\top \mathbf{Q} (\mathbf{B}^{-1/2}\mathbf{v}) > 0,$$

so it is PD.

Let $\tilde{\mathbf{g}}^{(k)} = \tilde{\mathbf{Q}}\mathbf{u}^{(k)} - \tilde{\mathbf{c}}$ be the gradient residual in the transformed coordinates. Then

$$\tilde{\mathbf{g}}^{(k)} = \mathbf{B}^{-1/2}\mathbf{g}^{(k)}.$$

If $\tilde{\mathbf{d}}^{(k)}$ is a CG direction in the transformed coordinates, set $\mathbf{d}^{(k)} = \mathbf{B}^{-1/2}\tilde{\mathbf{d}}^{(k)}$. The initial transformed direction gives

$$\mathbf{d}^{(0)} = -\mathbf{B}^{-1/2}\tilde{\mathbf{g}}^{(0)} = -\mathbf{B}^{-1}\mathbf{g}^{(0)} = -\mathbf{z}^{(0)}.$$

Also,

$$\begin{aligned} (\tilde{\mathbf{g}}^{(k)})^\top \tilde{\mathbf{g}}^{(k)} &= (\mathbf{g}^{(k)})^\top \mathbf{B}^{-1} \mathbf{g}^{(k)} = (\mathbf{g}^{(k)})^\top \mathbf{z}^{(k)}, \\ (\tilde{\mathbf{d}}^{(k)})^\top \tilde{\mathbf{Q}} \tilde{\mathbf{d}}^{(k)} &= (\mathbf{d}^{(k)})^\top \mathbf{Q} \mathbf{d}^{(k)}. \end{aligned}$$

Consequently, the ordinary-CG step length, conjugate coefficient, and direction formulas in the transformed coordinates become Equation (11.33), Equation (11.35), and Equation (11.36), respectively. The iterate and gradient transformations give Equation (11.34). Thus the transformed algorithm can be implemented by solving $\mathbf{B}\mathbf{z}^{(k)} = \mathbf{g}^{(k)}$, without forming $\mathbf{B}^{1/2}$ or $\mathbf{B}^{-1/2}$.

Finally, if $\tilde{\mathbf{e}}^{(k)} = \mathbf{u}^{(k)} - \mathbf{u}^* = \mathbf{B}^{1/2}\mathbf{e}^{(k)}$, then

$$\|\tilde{\mathbf{e}}^{(k)}\|_{\tilde{\mathbf{Q}}}^2 = (\mathbf{e}^{(k)})^\top \mathbf{Q} \mathbf{e}^{(k)} = \|\mathbf{e}^{(k)}\|_{\mathbf{Q}}^2.$$

If $\kappa_{\text{eff}}(\mathbf{B}) = 1$, the first case of Theorem 11.9 gives termination in at most one iteration. If $\kappa_{\text{eff}}(\mathbf{B}) > 1$, applying its second case to $\tilde{\mathbf{Q}}$ proves Equation (11.37) for every generated iterate. $\square$

Common choices include diagonal or Jacobi scaling, block-diagonal approximations, and preconditioners built from incomplete Cholesky factors. Relative to CG, PCG adds one solve $\mathbf{B}\mathbf{z}^{(k)} = \mathbf{g}^{(k)}$ per iteration; the single $\mathbf{Q}$ -vector product and short-recurrence storage are otherwise unchanged. A useful preconditioner balances the cost of this additional solve against the reduction in iteration count: applying $\mathbf{B}^{-1}$ should be inexpensive, while the eigenvalues of $\mathbf{B}^{-1/2}\mathbf{Q}\mathbf{B}^{-1/2}$ should be better clustered or have a smaller condition number. Positive definiteness of $\mathbf{B}$ alone does not guarantee improvement.

Warning

The derivation above assumes one fixed PD preconditioner $\mathbf{B}$. If the preconditioner changes with k , the standard PCG recurrences need not preserve the transformed-CG structure or conjugacy; flexible CG variants are designed for that setting.

11.7.1 A Jacobi-PCG comparison

The following example isolates a diagonal scaling imbalance that Jacobi preconditioning can remove cheaply. Let $n = 100$ and define

$$\begin{aligned} \mathbf{T} &= \text{tridiag}\left(-\frac{1}{4}, 1, -\frac{1}{4}\right), & \mathbf{D}_{ii} &= 10^{(i-1)/(n-1)}, \quad i = 1, \dots, n, \\ \mathbf{Q} &= \mathbf{D}\mathbf{T}\mathbf{D}. \end{aligned} \tag{11.38}$$

Set $\mathbf{x}^* = \mathbf{1}$, $\mathbf{c} = \mathbf{Q}\mathbf{1}$, and $\mathbf{x}^{(0)} = \mathbf{0}$. Since $\mathbf{T}$ has diagonal entries equal to one, its Jacobi preconditioner is

$$\mathbf{B} = \text{diag}(\mathbf{Q}) = \mathbf{D}^2.$$

Consequently,

$$\mathbf{B}^{-1/2}\mathbf{Q}\mathbf{B}^{-1/2} = \mathbf{D}^{-1}(\mathbf{D}\mathbf{T}\mathbf{D})\mathbf{D}^{-1} = \mathbf{T}.$$

The eigenvalues of $\mathbf{T}$ are

$$\lambda_j(\mathbf{T}) = 1 - \frac{1}{2} \cos\left(\frac{j\pi}{n+1}\right), \quad j = 1, \dots, n,$$

so $\mathbf{T} \succ 0$ and therefore $\mathbf{Q} = \mathbf{D}\mathbf{T}\mathbf{D} \succ 0$. Numerically,

$$\kappa(\mathbf{Q}) \approx 220.94, \quad \kappa_{\text{eff}}(\mathbf{B}) = \kappa(\mathbf{T}) \approx 2.998.$$

Using the relative energy-error target

$$\frac{\|\mathbf{x}^{(k)} - \mathbf{x}^*\|_{\mathbf{Q}}}{\|\mathbf{x}^{(0)} - \mathbf{x}^*\|_{\mathbf{Q}}} \leq 10^{-8},$$

the unpreconditioned and preconditioned runs give

method	condition number in its CG coordinates	updates
CG	220.94	88
Jacobi PCG	2.998	14

See also [Figure 11.3](#). This comparison illustrates the intended cost trade-off, rather than a universal advantage of Jacobi scaling. Here $\mathbf{B}$ cheaply removes the known diagonal scale variation. For a matrix whose difficulty is not primarily diagonal scaling, the same choice can provide little or no improvement.

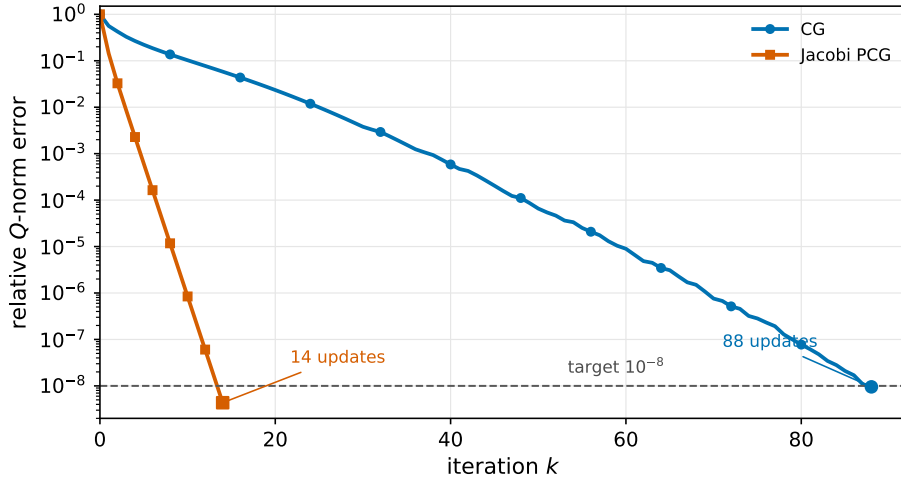


Figure 11.3: Relative Q -norm errors for CG and Jacobi PCG on Equation (11.38). The diagonal solve is only componentwise division, while the improved effective spectrum substantially reduces the number of updates.

11.8 Finite-precision residual refresh and restart

In exact arithmetic, the recursively updated gradient residual satisfies

$$\mathbf{g}^{(k)} = \mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}$$

at every iteration. In floating-point arithmetic, the two quantities can gradually drift apart: the recurrence accumulates all earlier rounding errors, whereas a fresh evaluation of $\mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}$ does not. A residual-refresh restart replaces

$$\mathbf{g}^{(k)} \leftarrow \text{fl}(\mathbf{Q}\mathbf{x}^{(k)} - \mathbf{c}), \quad \mathbf{d}^{(k)} \leftarrow -\mathbf{g}^{(k)}. \quad (11.39)$$

The first assignment repairs accumulated residual drift up to the rounding error of one new matrix–vector product; the second discards the contaminated direction recurrence. This repair is not free, because evaluating $\mathbf{Q}\mathbf{x}^{(k)}$ costs an additional matrix–vector product.

To display the loss of conjugacy, consider the 4×4 Hilbert matrix

$$Q_{ij} = \frac{1}{i+j-1}, \quad \mathbf{c} = \mathbf{Q}\mathbf{1}, \quad \mathbf{x}^{(0)} = \mathbf{0}, \quad \kappa(\mathbf{Q}) \approx 1.55 \times 10^4. \quad (11.40)$$

For the numerical experiment, $\mathbf{Q}$ and $\mathbf{c}$ are rounded separately to single precision and all CG operations use single precision. Let $\mathbf{x}_{\text{ref}}$ denote the solution obtained by interpreting this stored pair in double precision and solving it there. The reference is used only for diagnostics; all reported energy errors use $\mathbf{x}_{\text{ref}}$ and the stored $\mathbf{Q}$. This convention separates iteration error from the initial rounding of the problem data.

The normalized conjugacy defect of the newly generated direction is

$$\eta_k := \max_{0 \leq i < k} \frac{|(\mathbf{d}^{(i)})^\top \mathbf{Q} \mathbf{d}^{(k)}|}{\sqrt{(\mathbf{d}^{(i)})^\top \mathbf{Q} \mathbf{d}^{(i)}} \sqrt{(\mathbf{d}^{(k)})^\top \mathbf{Q} \mathbf{d}^{(k)}}}. \quad (11.41)$$

Exact arithmetic gives $\eta_k = 0$. For the stored Hilbert system, the single-precision values for the directions $\mathbf{d}^{(1)}, \mathbf{d}^{(2)}, \mathbf{d}^{(3)}$ are approximately 5.60×10^{-6} , 2.39×10^{-2} , and 0.9996 ,

respectively. Thus the fourth direction is almost parallel, in the $\mathbf{Q}$ -inner-product geometry, to one of the earlier directions. The corresponding double-precision defects remain much smaller over these first four directions.

For a clearer display of what a refresh can repair, consider a structured 20-dimensional system. Let

$$\begin{aligned} T &= \text{tridiag}(-1, 2, -1), & D_{ii} &= 10^{(i-1)/19}, \\ \mathbf{Q} &= \frac{DTD}{\lambda_{\max}(DTD)}, & \mathbf{c} &= \mathbf{Q}\mathbf{e}_1, & \mathbf{x}^{(0)} &= \mathbf{0}. \end{aligned} \quad (11.42)$$

The matrix is formed in double precision, normalized, and then stored in single precision; $\mathbf{c}$ is set equal to the first column of the stored matrix. Hence the reference solution of the stored system is exactly $\mathbf{x}_{\text{ref}} = \mathbf{e}_1$. The stored matrix is symmetric PD and has $\kappa(\mathbf{Q}) \approx 2.402 \times 10^3$.

Compare ordinary CG with a run that applies Equation (11.39) after every 30 CG updates. Each refresh product is charged to a common budget. Over 100 matrix–vector products, ordinary CG stagnates, whereas the restarted run passes through three residual refreshes and reaches a much lower error. At the final common budget, the measured relative energy errors are

method	$\ \mathbf{x}^{(k)} - \mathbf{x}_{\text{ref}}\ _{\mathbf{Q}} / \ \mathbf{x}^{(0)} - \mathbf{x}_{\text{ref}}\ _{\mathbf{Q}}$
ordinary CG	7.878399×10^{-8}
refresh and restart every 30 updates	1.039277×10^{-9}

The ordinary recurrence reports a relative residual of only 3.84×10^{-20} , even though direct recomputation gives 7.52×10^{-8} . This is the residual gap. For the restarted run, the corresponding values are 8.49×10^{-10} and 1.20×10^{-9} , so the recursive residual again reflects the actual accuracy.

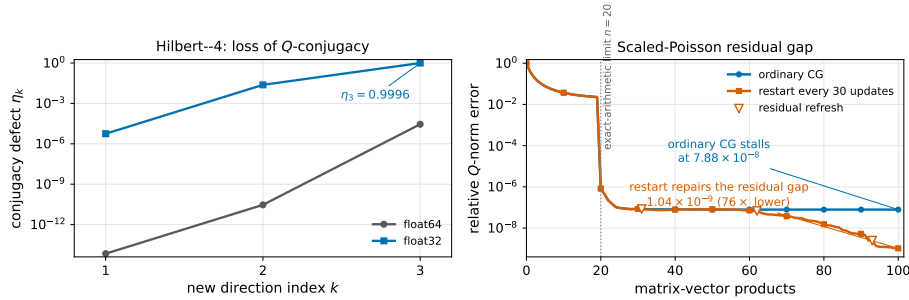


Figure 11.4: Two finite-precision effects. The left panel uses the Hilbert system in Equation (11.40) and plots Equation (11.41), directly showing the loss of conjugacy in single precision. The right panel uses the scaled-Poisson system in Equation (11.42): ordinary CG plateaus, while residual refresh and restart reach an error about 76 times smaller. The horizontal axis charges both ordinary $\mathbf{Q}\mathbf{d}^{(k)}$ products and the extra $\mathbf{Q}\mathbf{x}^{(k)}$ products used for refreshes.

Warning

The improvement in Figure 11.4 is a finite-precision effect, not a stronger exact-arithmetic property. Exact linear CG already retains all previous Krylov information and terminates in at most n updates. Restarting in exact arithmetic discards some of that information and cannot improve this ideal finite-termination guarantee. Nor

is a fixed restart period uniformly beneficial in floating point: a restart that is too frequent can spend extra products while repeatedly discarding useful directions.

11.9 From linear CG to nonlinear conjugate gradients

For a general differentiable objective $f : \mathbb{R}^n \rightarrow \mathbb{R}$, there is no fixed matrix $\mathbf{Q}$ defining one global quadratic geometry. Nevertheless, one can retain the inexpensive two-term direction recurrence. These methods are called *nonlinear conjugate-gradient methods*.

11.9.1 The generic nonlinear recurrence

Set

$$\mathbf{g}^{(k)} = \nabla f(\mathbf{x}^{(k)}), \quad \mathbf{d}^{(0)} = -\mathbf{g}^{(0)},$$

and, for $k \geq 1$, use

$$\mathbf{d}^{(k)} = -\mathbf{g}^{(k)} + \beta^{(k)} \mathbf{d}^{(k-1)}, \quad \mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}. \quad (11.43)$$

Exact line search is still mathematically meaningful whenever the line problem has a minimizer, but it usually has no closed form and can itself be a difficult global one-dimensional problem. Practical nonlinear CG therefore uses an inexact line search.

For a descent direction, for example, the strong Wolfe conditions choose $0 < c_1 < c_2 < 1$ and require

$$f(\mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)}) \leq f(\mathbf{x}^{(k)}) + c_1 \alpha^{(k)} (\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)}, \quad (11.44)$$

$$\left| \nabla f(\mathbf{x}^{(k)} + \alpha^{(k)} \mathbf{d}^{(k)})^\top \mathbf{d}^{(k)} \right| \leq c_2 \left| (\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)} \right|. \quad (11.45)$$

The first inequality is Armijo sufficient decrease, while the second prevents the line search from stopping too early.

11.9.2 Classical conjugate parameters

Let

$$\mathbf{y}^{(k-1)} := \mathbf{g}^{(k)} - \mathbf{g}^{(k-1)}, \quad k \geq 1.$$

Four classical choices are

$$\beta_{\text{FR}}^{(k)} = \frac{\|\mathbf{g}^{(k)}\|^2}{\|\mathbf{g}^{(k-1)}\|^2}, \quad (11.46)$$

$$\beta_{\text{PRP}}^{(k)} = \frac{(\mathbf{g}^{(k)})^\top \mathbf{y}^{(k-1)}}{\|\mathbf{g}^{(k-1)}\|^2}, \quad \beta_{\text{PRP}+}^{(k)} = \max\{\beta_{\text{PRP}}^{(k)}, 0\}, \quad (11.47)$$

$$\beta_{\text{HS}}^{(k)} = \frac{(\mathbf{g}^{(k)})^\top \mathbf{y}^{(k-1)}}{(\mathbf{d}^{(k-1)})^\top \mathbf{y}^{(k-1)}}, \quad (11.48)$$

$$\beta_{\text{DY}}^{(k)} = \frac{\|\mathbf{g}^{(k)}\|^2}{(\mathbf{d}^{(k-1)})^\top \mathbf{y}^{(k-1)}}. \quad (11.49)$$

Here FR denotes Fletcher–Reeves, PRP denotes Polak–Ribière–Polyak, HS denotes Hestenes–Stiefel, and DY denotes Dai–Yuan. A formula with a zero or numerically tiny denominator must not be used without a safeguard.

Proposition 11.11 (Agreement on a PD quadratic). *Apply the nonlinear recurrence to the quadratic Equation (11.1), use exact line search, and suppose that the iterates agree with linear CG through an iteration $k \geq 1$. Before termination,*

$$\beta_{\text{FR}}^{(k)} = \beta_{\text{PRP}}^{(k)} = \beta_{\text{PRP}+}^{(k)} = \beta_{\text{HS}}^{(k)} = \beta_{\text{DY}}^{(k)}. \quad (11.50)$$

Therefore each of these formulas reproduces the linear-CG direction in exact arithmetic.

Proof. Linear CG gives $(\mathbf{g}^{(k)})^\top \mathbf{g}^{(k-1)} = 0$ by Theorem 11.6. Consequently,

$$(\mathbf{g}^{(k)})^\top \mathbf{y}^{(k-1)} = \|\mathbf{g}^{(k)}\|^2. \quad (11.51)$$

Exact line search at iteration $k-1$ gives $(\mathbf{g}^{(k)})^\top \mathbf{d}^{(k-1)} = 0$, whereas Equation (11.20) gives $(\mathbf{g}^{(k-1)})^\top \mathbf{d}^{(k-1)} = -\|\mathbf{g}^{(k-1)}\|^2$. Hence

$$(\mathbf{d}^{(k-1)})^\top \mathbf{y}^{(k-1)} = \|\mathbf{g}^{(k-1)}\|^2. \quad (11.52)$$

Substitution of Equations (11.51) and (11.52) into Equations (11.46) to (11.49) shows that every coefficient equals $\|\mathbf{g}^{(k)}\|^2 / \|\mathbf{g}^{(k-1)}\|^2$. This number is nonnegative, so PRP+ also has the same value. All variants start from the common direction $\mathbf{d}^{(0)} = -\mathbf{g}^{(0)}$; induction on k therefore shows that each formula reproduces the linear-CG direction and iterate until termination. $\square$

11.9.3 What is lost, and how restarts help

For a nonquadratic objective, the Hessian changes from point to point. Thus there is no single matrix with respect to which all directions remain conjugate, the Krylov-subspace identities no longer apply, and finite termination is lost. Moreover, none of the formulas Equations (11.46) to (11.49) by itself guarantees that $\mathbf{d}^{(k)}$ is a descent direction for every general objective and every inexact line search.

A simple safeguard tests the direction after it is formed. Fix $\eta \in (0, 1)$ and restart with steepest descent whenever

$$(\mathbf{g}^{(k)})^\top \mathbf{d}^{(k)} > -\eta \|\mathbf{g}^{(k)}\|^2; \quad \text{then set } \beta^{(k)} = 0, \quad \mathbf{d}^{(k)} = -\mathbf{g}^{(k)}. \quad (11.53)$$

The accepted direction consequently satisfies a quantitative descent condition. Periodic restarts and restarts triggered by a loss of gradient orthogonality are also common. PRP+ is often effective in computations because a negative PRP coefficient automatically restarts the method, but this empirical observation is not a universal dominance theorem.

The restart test therefore restores the basic premise needed by Armijo backtracking. Stronger global convergence conclusions for a particular nonlinear-CG variant require additional assumptions on f and on the line search; the linear-CG finite-termination and Chebyshev results must not be transferred to the nonlinear method.

Chapter summary

- Conjugate directions give exact line search a memory: after k steps, the iterate minimizes the quadratic over the affine span of all k directions.
- Nonzero $\mathbf{Q}$ -conjugate directions are independent, so a complete conjugate basis reaches the minimizer in at most n exact-arithmetic steps.
- CG constructs the next direction from the current negative gradient and only the previous direction. Its directions are mutually $\mathbf{Q}$ -conjugate, its gradients are mutually orthogonal, and both families span the same Krylov subspaces.
- The CG error bound depends on $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$, improving the worst-case condition- number dependence of steepest descent.
- PCG applies CG in the geometry of $\mathbf{B}^{-1/2}\mathbf{Q}\mathbf{B}^{-1/2}$; the solve with $\mathbf{B}$ must be cheap enough to justify the improved effective spectrum.
- In finite precision, a residual-refresh restart can repair accumulated recurrence drift, but it spends an extra matrix-vector product and does not improve exact-arithmetic linear CG.
- Nonlinear CG retains the two-term recurrence but generally loses exact conjugacy, finite termination, and automatic descent. FR, PRP, HS, and DY agree on a PD quadratic with exact line search but differ on general objectives; line-search and restart safeguards matter.

11.10 Exercises

The exercises are ordered from direct calculations to proof-based and spectral questions. Their numerical data are different from the examples used in the chapter and in the slides.

Exercise 11.1 (Constructing a conjugate direction). Let

$$\mathbf{Q} = \begin{pmatrix} 4 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{d}^{(0)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

Find a nonzero vector $\mathbf{d}^{(1)}$ that is $\mathbf{Q}$ -conjugate to $\mathbf{d}^{(0)}$. Verify that your two directions need not be Euclidean- orthogonal.

Exercise 11.2 (One exact line search). Let

$$q(\mathbf{x}) = \frac{1}{2}\mathbf{x}^\top \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix} \mathbf{x} - \begin{pmatrix} 1 \\ 0 \end{pmatrix}^\top \mathbf{x}, \quad \mathbf{x}^{(0)} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \mathbf{d}^{(0)} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Compute $\mathbf{g}^{(0)}$, the exact step length, $\mathbf{x}^{(1)}$, and $\mathbf{g}^{(1)}$. Verify $(\mathbf{g}^{(1)})^\top \mathbf{d}^{(0)} = 0$.

Exercise 11.3 (Gradient residual and stopping). Let

$$\mathbf{Q} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} 0.3 \\ 0.8 \end{pmatrix}.$$

Compute the gradient residual and the conventional linear-system residual. If $\varepsilon_{\text{abs}} = 10^{-8}$ and $\varepsilon_{\text{rel}} = 10^{-4}$, determine whether Equation (11.23) is satisfied.

Exercise 11.4 (Two conjugate line searches). For

$$\mathbf{Q} = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{x}^{(0)} = \mathbf{0},$$

use $\mathbf{d}^{(0)} = (1, 0)^\top$ and $\mathbf{d}^{(1)} = (1, -3)^\top$. Verify conjugacy, perform both exact line searches, and check that the final point solves $\mathbf{Q}\mathbf{x} = \mathbf{c}$.

Exercise 11.5 (A fresh two-dimensional CG calculation). Apply CG from $\mathbf{x}^{(0)} = \mathbf{0}$ to

$$\mathbf{Q} = \begin{pmatrix} 4 & 1 \\ 1 & 3 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Compute every $\mathbf{g}^{(k)}$, $\mathbf{d}^{(k)}$, $\alpha^{(k)}$, $\beta^{(k)}$, and $\mathbf{x}^{(k)}$ until termination. Verify gradient orthogonality and $\mathbf{Q}$ -conjugacy directly.

Exercise 11.6 (Using and comparing rate bounds). Suppose $\kappa(\mathbf{Q}) = 81$. Use Equation (11.26) to find a sufficient number of CG iterations for $\|\mathbf{e}^{(k)}\|_{\mathbf{Q}} \leq 10^{-6} \|\mathbf{e}^{(0)}\|_{\mathbf{Q}}$. Do the same using the exact-search steepest-descent factor $(\kappa - 1)/(\kappa + 1)$. Explain why neither sufficient count predicts the exact stopping iteration.

Exercise 11.7 (A Jacobi PCG calculation). Let

$$\mathbf{Q} = \begin{pmatrix} 100 & 9 \\ 9 & 1 \end{pmatrix}, \quad \mathbf{B} = \text{diag}(\mathbf{Q}).$$

Verify that both matrices are PD. Compute $\mathbf{B}^{-1/2}\mathbf{Q}\mathbf{B}^{-1/2}$ and its condition number, and compare it with $\kappa(\mathbf{Q})$. State what the comparison predicts for the CG bounds and what extra per-iteration work PCG requires.

Exercise 11.8 (Nonlinear-CG coefficients and descent). Suppose

$$\mathbf{g}^{(k-1)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \mathbf{g}^{(k)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \mathbf{d}^{(k-1)} = \begin{pmatrix} -1 \\ 4 \end{pmatrix}.$$

Compute $\beta_{\text{FR}}^{(k)}$, $\beta_{\text{PRP}}^{(k)}$, $\beta_{\text{PRP}+}^{(k)}$, $\beta_{\text{HS}}^{(k)}$, and $\beta_{\text{DY}}^{(k)}$. For each coefficient, form Equation (11.43) and determine whether the resulting direction is descending at $\mathbf{x}^{(k)}$. Apply the restart test Equation (11.53) with $\eta = 10^{-2}$.

Exercise 11.9 (Order independence of a complete conjugate basis). Let $\mathbf{d}^{(0)}, \dots, \mathbf{d}^{(n-1)}$ be a $\mathbf{Q}$ -conjugate basis. Prove that the coefficient of $\mathbf{d}^{(i)}$ in the expansion of $\mathbf{x}^* - \mathbf{x}^{(0)}$ is

$$-\frac{(\mathbf{g}^{(0)})^\top \mathbf{d}^{(i)}}{(\mathbf{d}^{(i)})^\top \mathbf{Q} \mathbf{d}^{(i)}}.$$

Deduce that exact line searches reach the same final point in any ordering of the basis,

even though their intermediate iterates can differ.

Exercise 11.10 (Early termination from the active spectrum, advanced). Assume that the initial error $e^{(0)}$ has nonzero components in exactly s distinct eigenspaces of Q . Construct a polynomial p of degree s satisfying $p(0) = 1$ and $p(\lambda) = 0$ at those s eigenvalues. Use [Lemma 11.8](#) to prove that CG terminates in at most s iterations in exact arithmetic.